DSA ALGORITHMS TOOLKIT

Project Report — Python

Project type: Data Structures and Algorithms mini project
Language: Python 3
Platform: Windows, macOS, or Linux terminal

1. Abstract

The DSA Algorithms Toolkit is a menu-driven Python application designed to help beginners practise core Data Structures and Algorithms concepts. It implements linear search, binary search, bubble sort, selection sort, insertion sort, balanced-bracket checking using a stack, and duplicate removal. The program accepts user input and prints the result of the selected operation.

2. Introduction

Data Structures and Algorithms are fundamental to programming. Data structures organise information, while algorithms describe steps to solve problems. Learning their behavior and time complexity helps students choose suitable approaches for real applications.

3. Objectives

• Practise common searching and sorting algorithms.
• Understand a stack through balanced-bracket checking.
• Learn how a set can help identify duplicate values.
• Display results through a simple command-line menu.
• Introduce time and space complexity.

4. Requirements

Software: Python 3.9 or newer and a terminal or code editor. No third-party packages are needed.
Hardware: Any computer that can run Python.

5. Modules

Searching: Linear Search checks values one by one. Binary Search repeatedly halves a sorted search range.
Sorting: Bubble Sort swaps neighboring values, Selection Sort repeatedly selects the minimum, and Insertion Sort inserts each value into the already-sorted section.
Stack: Balanced-bracket checking pushes opening brackets and matches them with closing brackets.
Duplicate removal: A set tracks values already seen while the output list preserves first-seen order.

6. Complexity Analysis

Linear Search: O(n) average time. Binary Search: O(log n) time on sorted input. Bubble Sort, Selection Sort, and Insertion Sort: $O(n^2)$ average time. Balanced Brackets: O(n) time and O(n) extra space. Duplicate removal: O(n) average time with a set. The Binary Search menu option sorts the input first, so its complete menu operation also includes sorting cost.

7. How to Run

1. Install Python 3.9 or newer.
2. Open the project folder in VS Code or a terminal.

3. Run `python main.py` (or `python3 main.py`).
4. Choose a menu option and enter the requested values.

8. Sample Output

```
DSA ALGORITHMS TOOLKIT
Choose an option: 3
Enter integers separated by spaces: 8 3 5 1
Sorted: [1, 3, 5, 8]
```

9. Advantages and Limitations

Advantages: Beginner-friendly, interactive, uses no external packages, and demonstrates several foundational algorithms.
Limitations: It is a learning toolkit, not a production library. It does not yet include trees, graphs, linked lists, or visual animations.

10. Future Scope

Future versions can add linked lists, queues, hash tables, trees, graph traversal, shortest-path algorithms, automated tests, and visual step-by-step demonstrations.

11. Conclusion

The DSA Algorithms Toolkit combines several essential beginner algorithms in one small Python project. It provides a practical way to practise algorithm logic and begin understanding complexity.

12. References

1. Python documentation: https://docs.python.org/3/
2. Course notes on Data Structures and Algorithms.
3. Standard algorithm analysis concepts commonly taught in introductory computer science.